

Tools & Libraries — Expanded Practical Guide

Frameworks and software useful for implementing and evaluating multi-tool agentic AI systems

1. Tool selection criteria

The assignment asks for useful software/frameworks with brief descriptions. For this topic, a tool is most useful when it helps construct an agent, expose tools, implement routing, capture traces, or systematically evaluate behavior.

1. LangGraph

Purpose: Stateful agent orchestration and graph-based workflows.

Use in this project: Implement explicit routing nodes, candidate-tool selection, state transitions and evaluation traces.

Official project: <https://www.langchain.com/langgraph>

2. LangChain

Purpose: Framework for LLM applications, tools, agents and retrieval.

Use in this project: Build baseline multi-tool agents and compare routing strategies.

Official project: <https://www.langchain.com/>

3. Microsoft AutoGen

Purpose: Framework for multi-agent and tool-enabled applications.

Use in this project: Useful for studying tool choice when multiple agents collaborate.

Official project: <https://github.com/microsoft/autogen>

4. LlamaIndex

Purpose: Framework for agentic applications and data/tool integration.

Use in this project: Useful for agents selecting tools over data and services.

Official project: <https://www.llamaindex.ai/>

5. DSPy

Purpose: Framework for programming and optimizing LM-based pipelines.

Use in this project: Useful for systematic prompt/module optimization and evaluation experiments.

Official project: <https://dspy.ai/>

2. Suggested experimental architecture

Use a small controller layer around an agent framework. The controller receives a task, constructs a candidate-tool list, invokes the agent, records the selected tool and arguments, executes the tool in a safe test environment, and stores an evaluation record. This separation makes it possible to score selection independently from final-answer quality.

3. Trace fields

Trace field	Purpose
task	Original user instruction

Trace field	Purpose
tools_available	Candidate tool set
selected_tool	Agent decision
arguments	Generated parameters
tool_output	Observed result
final_answer	Agent response
expected_tool	Reference decision
score	Evaluation result

4. Tool comparison questions

- Can the framework expose multiple candidate tools?
- Can tool selection be logged separately from execution?
- Can multi-step state be preserved?
- Can experiments be repeated with controlled tool descriptions?
- Can evaluation traces be exported for analysis?